Fault Detection in UAVs: A Binary Black-Box Classification Approach

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Abstract

This work addresses the problem of Fault Detection and Isolation (FDI) in Unmanned Aerial Vehicles (UAVs) by leveraging the RflyMAD dataset. We propose a supervised, binary classification approach that, despite the sequential nature of flight data, operates on static instantaneous records. This hypothesis posits that the instantaneous state of the system, represented by a multidimensional feature vector, contains sufficient discriminative information to determine the aircraft's health status without relying on long-term temporal memory. We present a comprehensive data processing workflow designed to transform continuous telemetry streams into structured tabular datasets suitable for "Black Box" machine learning algorithms. The study evaluates the efficacy of these models in identifying various failure modes, serving as a foundation for robust predictive maintenance strategies.

1 Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly critical in diverse sectors, from logistics to surveillance. Ensuring their operational reliability is paramount, necessitating effective Fault Detection and Isolation (FDI) systems. This paper explores a data-driven approach to FDI, utilizing the extensive RflyMAD dataset, which integrates high-fidelity simulation data (SIL/HIL) with real flight records.

While flight data is inherently a time series, this study investigates the viability of a static classification paradigm. We hypothesize that the correlation between system variables at any given instant captures the "footprint" of a fault (e.g., a discrepancy between control signals and inertial response), enabling detection without complex temporal modelling. This simplification allows for the deployment of classic but powerful classification algorithms that classify the system state based on immediate sensor readings.

The RflyMAD dataset serves as an ideal foundation for this research due to its volume and heterogeneity. It encompasses thousands of high-fidelity simulation scenarios and hundreds of real-world flights, providing a dense feature space where decision boundaries can be rigorously tested. From the perspective of predictor variables, the dataset offers high-frequency readings from accelerometers, gyroscopes, magnetometers, and control signals. This diversity is crucial for building a model that is robust across different flight contexts.

By shifting the focus to isolated records, the analytical challenge transforms from sequence modelling to the robust representation of individual instances. The model is tasked with identifying fault conditions ranging from motor failures to sensor errors, solely based on the "statistical photograph" of the sensors at a specific moment. Consequently, this document outlines a workflow for feature engineering and the construction of a tabular dataset, detailing the transformation of raw logs into viable training records through segmentation, synchronisation, and statistical extraction.

2 State of the art for predictive maintenance

Predictive Maintenance (PdM) has evolved from schedule-based interventions to Condition-Based Maintenance (CBM), driven by the availability of sensor data and advanced analytics. Data-driven approaches fall into three broad categories: statistical models, Machine Learning (ML), and Deep Learning (DL) methods.

Traditional ML techniques, such as Support Vector Machines (SVM) (Widodo and Yang, 2007) and Random Forests (RF), have been widely applied for fault diagnosis (Lei et al., 2018). These "Grey Box" or "White Box" models offer a balance between performance and interpretability.

In recent years, "Black Box" models, particularly Deep Learning architectures, have gained prominence due to their ability to model complex non-linear relationships in high-dimensional data (Zhang et al., 2019). Carvalho et al. (2019) highlight the growing trend of using Artificial Neural Networks (ANNs) and Deep Neural Networks (DNNs) for PdM, noting their superior performance in feature extraction and classification accuracy. For instance, Li et al. (2019) and Zhao et al. (2019) demonstrate the effectiveness of Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) in learning fault signatures from raw sensor data. Moreover, Hinton and Salakhutdinov (2006) pioneered methods for dimensionality reduction using neural networks, which underpin modern autoencoder-based anomaly detection.

Despite their opacity, these Black Box models often outperform traditional methods when dealing with vast and heterogeneous datasets like RflyMAD, provided that robust feature engineering and adequate regularisation are applied (Bishop, 2006; Susto et al., 2015).

3 Data processing domain

3.1 Data collection

The ingestion of raw data is managed through a transformation pipeline that converts unstructured telemetry into manageable analytical assets. Initially, official Data Processing Tools are used to process binary flight logs, extracting and synchronising sensor signals into intermediate CSV files for each flight. To handle the massive volume of data, exceeding conventional memory limits, a consolidation stage concatenates thousands of dispersed CSV files from SIL, HIL, and Real subdomains into Apache Parquet format. This columnar storage enables efficient out-of-core processing using Dask, facilitating parallel manipulation without saturating system RAM.

3.2 Data adequacy

Data consolidation and cleaning were performed to ensure structural consistency across HIL, SIL, and Real datasets. To optimise analysis under Tidy Data principles, new indicator variables were generated to segment examples by operational nature and flight conditions (Kotsiantis et al., 2006). Binary variables were created to identify specific faults (e.g., `Fault_Type_motor`, `Fault_Type_GPS`) and flight states (e.g., `Flight_Status_hover`). Metadata variables such as `Source_File` were included to maintain traceability.

3.3 Data split

A rigorous splitting strategy was employed to prevent data leakage. A random split of individual rows would fragment time series from the same flight into both training and test sets, leading to over-optimistic performance as the model would merely interpolate trajectories. Instead, we split data based on complete flight series (by Flight ID). This ensures that validation and test sets contain entirely new flights never seen by the model during training. A 50/50 split was chosen: 50% for training and 50% for testing, leveraging the law of large numbers to ensure statistical representativeness given the massive dataset size.

3.4 Data cleansing

3.4.1 Inconsistent data

We verified semantic integrity across datasets. In the Real dataset, specific corrupted files (e.g., `Case_3200000035.csv`) and cases with discrepancies between reported fault types and internal codes (e.g., GPS faults showing Magnetometer error codes) were identified and removed to avoid introducing label noise.

3.4.2 Missing values and static columns

Columns with single constant values (zero variance) were identified and removed. For missing values (NaN), given the millions of available records, we opted for a conservative approach: dropping columns with null values rather than imputing them, prioritising data veracity over quantity.

3.4.3 Outliers

Outliers were identified using Interquartile Range (IQR), Z-Score, and Isolation Forest. The analysis highlighted a higher density of outliers during fault states. Figure 1 shows the distribution of outliers across key sensor variables.

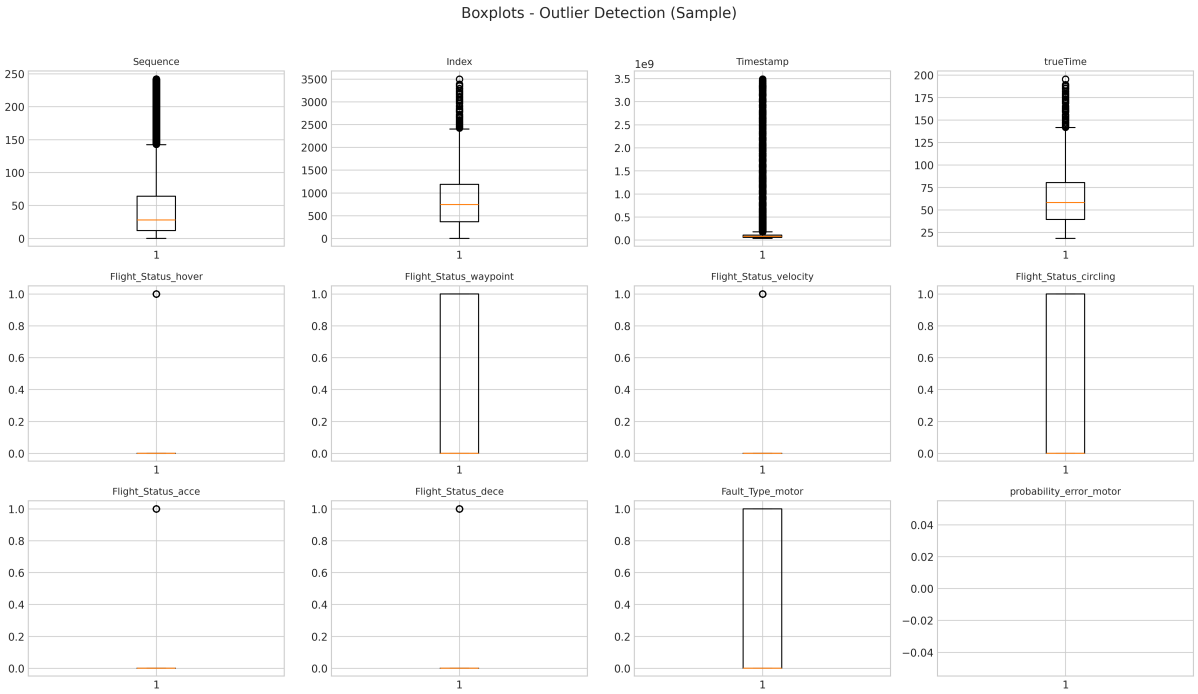


Figure 1: Boxplots showing the distribution and outliers for key variables.

4 Exploratory data analysis

The EDA phase focused on the "clean" dataset to characterise sensor behaviour. Univariate analysis (histograms/KDE) checked for Gaussianity, while correlation matrices identified multicollinearity. Figure 2 presents the distribution of selected variables.

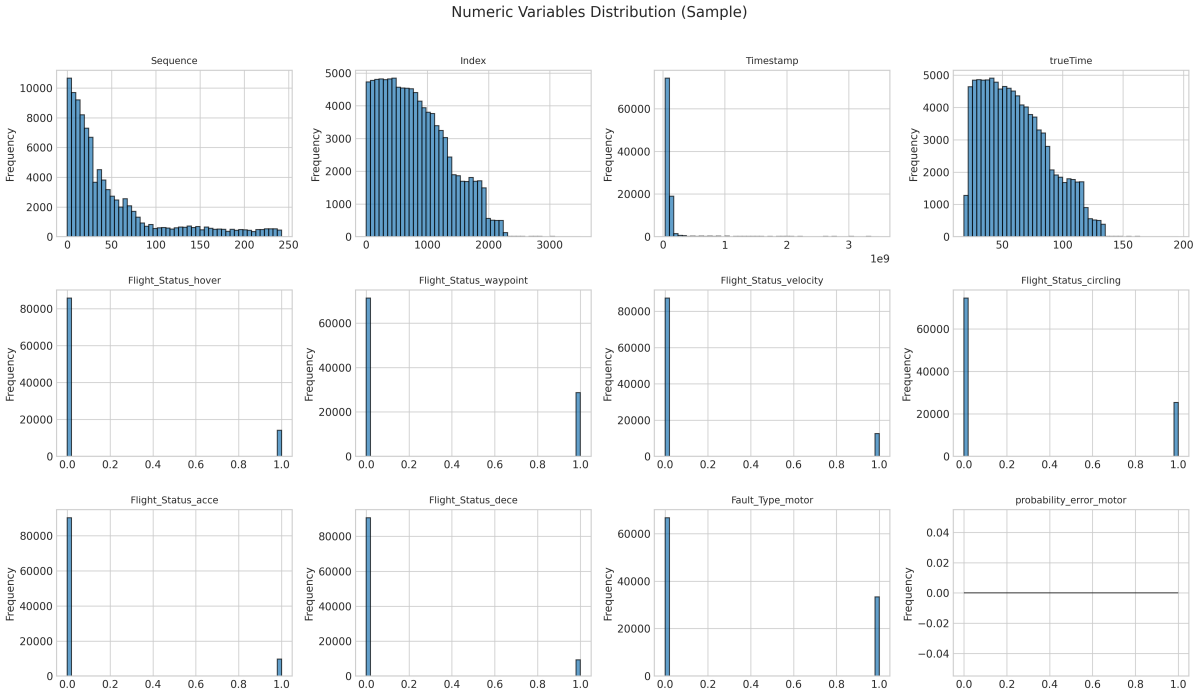


Figure 2: Histograms and density plots for selected feature variables.

The relationship between variables was analysed to detect redundancy. High correlations between certain physical sensors suggest that dimensionality reduction techniques like PCA could be effective (Figure 3).

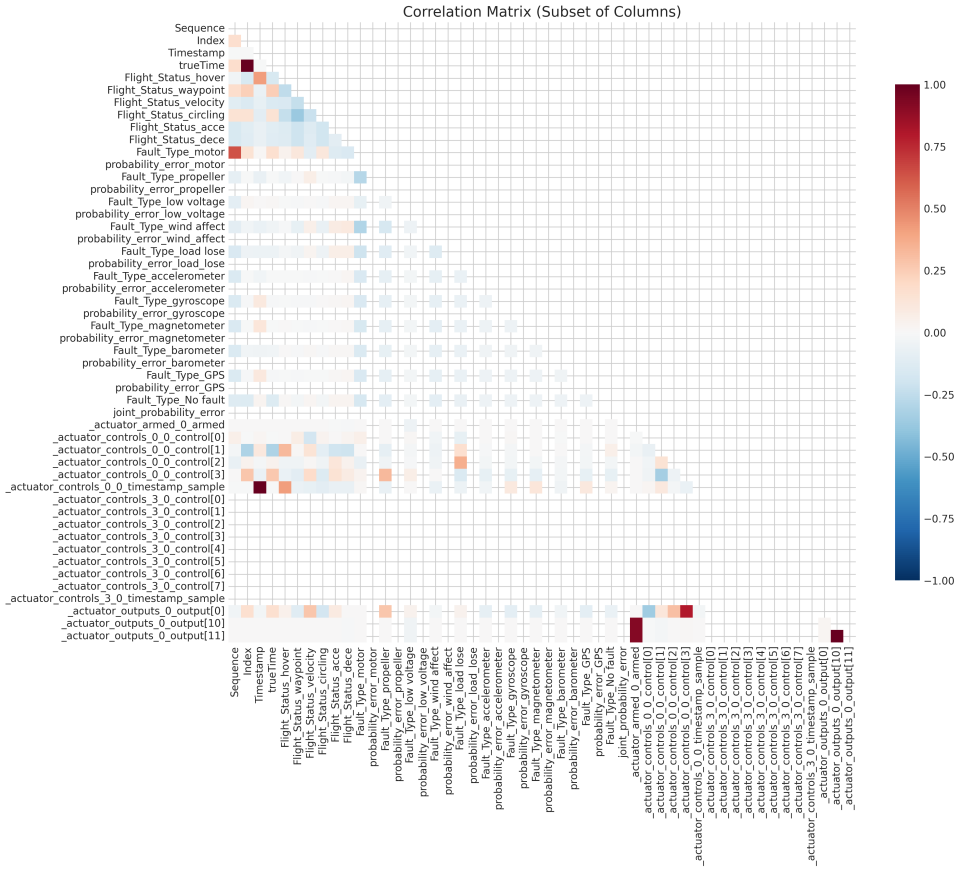


Figure 3: Correlation matrix of the features.

4.1 Data balancing analysis

An analysis of class distribution revealed a significant imbalance. In simulation (HIL/SIL), motor faults predominate (Figure 4, 5). The Real dataset is smaller and lacks certain fault types like “low_voltage” or “propeller” faults (Figure 6). This imbalance necessitates the use of metrics like F1-Score rather than accuracy.

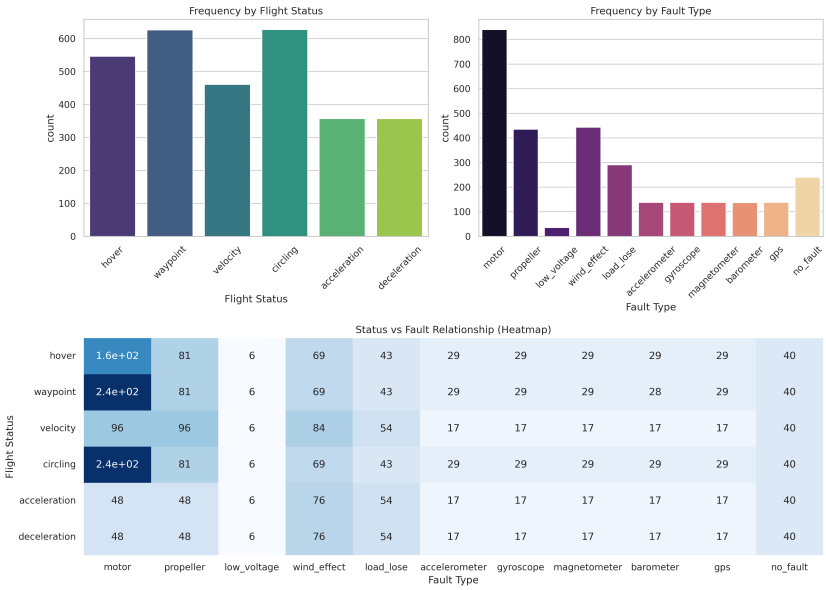


Figure 4: HIL distribution

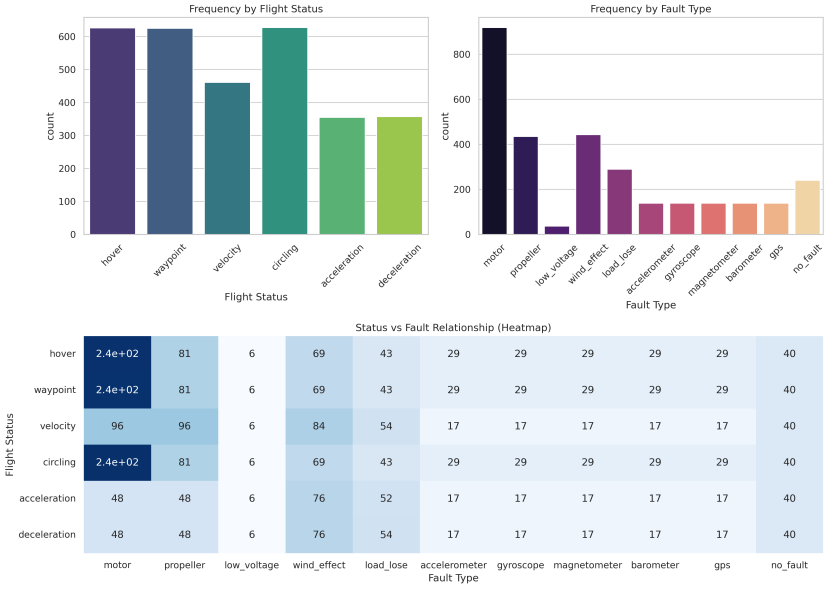


Figure 5: SIL distribution

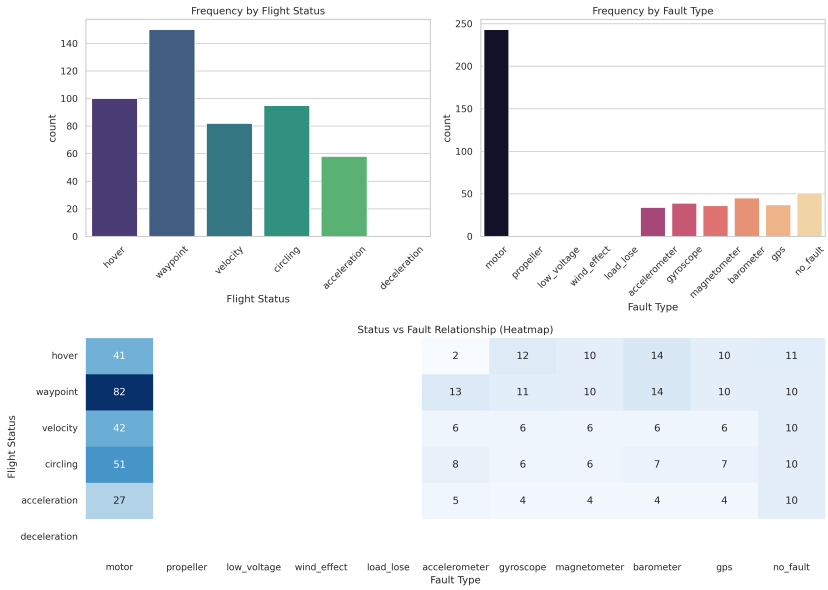


Figure 6: Real distribution

5 Features engineering domain

5.1 Feature data transform

Since no categorical predictors were present, only numerical transformation was required. We applied `StandardScaler` to standardise all features to zero mean and unit variance ($N(0, 1)$), which is critical for the performance of gradient-based algorithms like Neural Networks.

5.2 Feature Importance and Selection

Feature importance analysis using Random Forest revealed potential biases. Variables such as `program_time` or setpoint altitudes were found to be artificially correlated with faults due to the rigid testing protocols (e.g., faults always occurring at the same mission step). These "leaky" variables were manually removed to force the model to learn from flight dynamics rather than experimental artifacts. Figure 7 shows the importance of features before final selection.

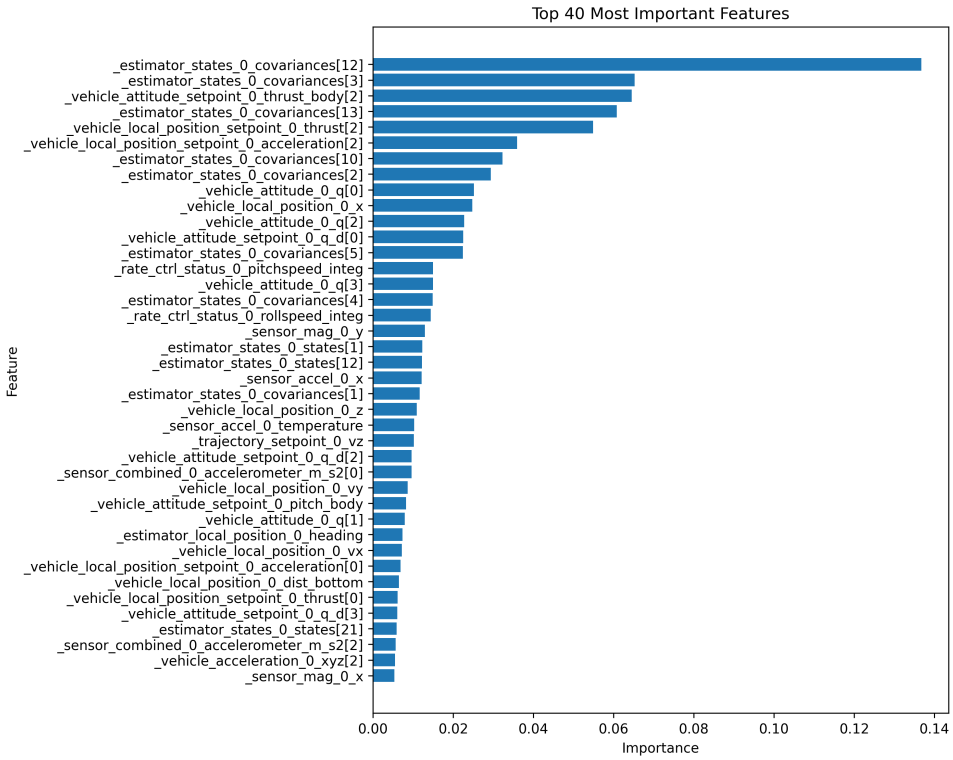


Figure 7: Feature importance scores derived from Random Forest.

After manual removal of biased features and filtering for multicollinearity (Guyon and Elisseeff, 2003), a refined set of variables remained.

5.3 Feature extraction

Principal Component Analysis (PCA) was applied to the cleaned feature set to reduce dimensionality while maintaining variance (Jolliffe, 2002). Analysis showed that 53 components are required to retain 95% of the variance (Figure 8). This transformation provides orthogonal, uncorrelated features to the classifiers.

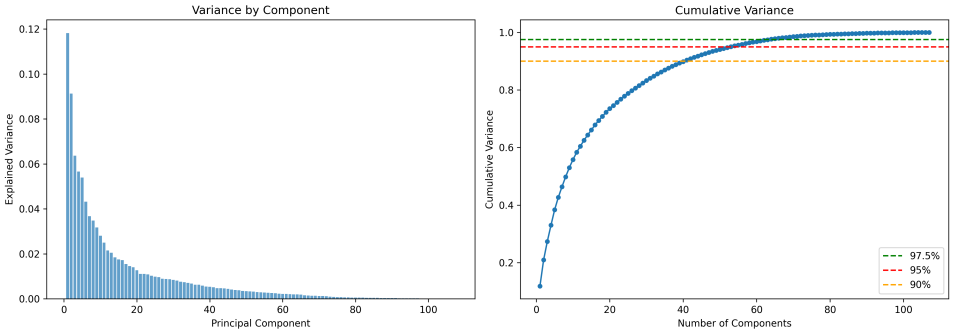


Figure 8: Cumulative variance explained by Principal Components.

The projection of data onto the first two components is visualized in Figure 9.

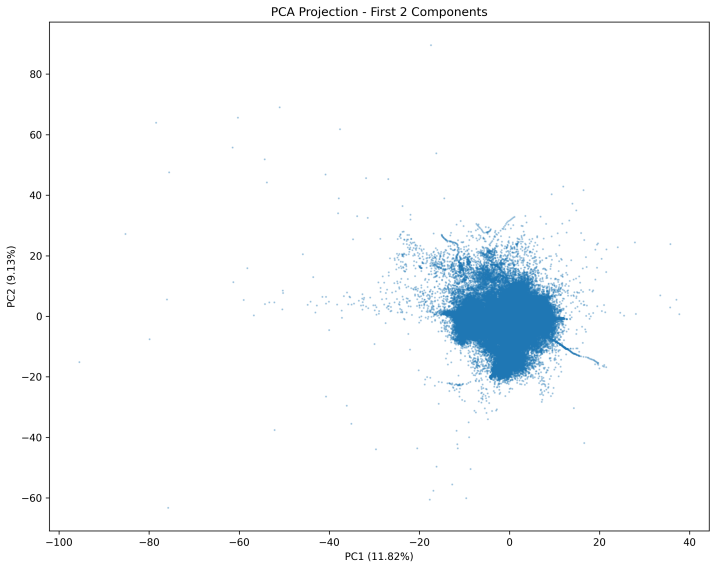


Figure 9: 2D Projection of the data using the first two principal components.

6 Model development domain

The predictive strategy focuses on binary classification (fault vs. nominal). Two experimental paths were established for feature input:

- **Domain knowledge selection**: A manually selected subset of 13 critical kinematic variables.
- **PCA Projection**: A set of principal components retaining 90% of the variance.

In this work, we prioritise "Black Box" models, specifically Multi-Layer Perceptrons (MLP). Given the class imbalance, optimisation focused on F1-Score and recall rather than accuracy.

The evaluation of Black Box models (MLP) demonstrates robust performance in detecting faults. Figure 10 illustrates the F1-Scores obtained, showing how MLP models perform competitively, particularly when dimensionality is reduced via PCA.

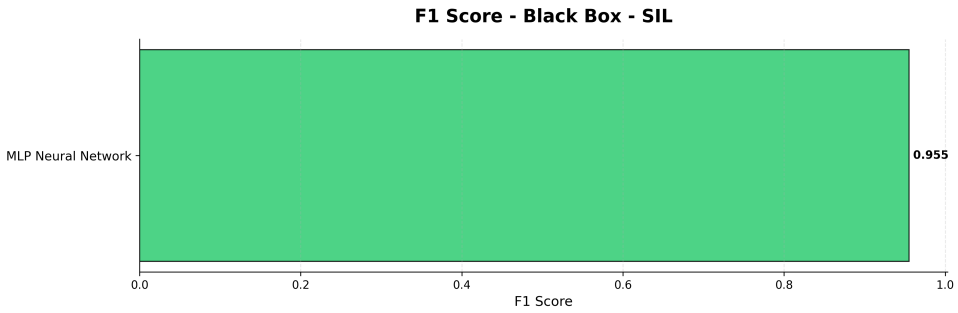


Figure 10: F1 Scores of the evaluated models (Black Box focus).

The confusion matrices (Figure 11) reveal that the models successfully isolate the minority class (faults) with few false negatives, which is critical for safety data.

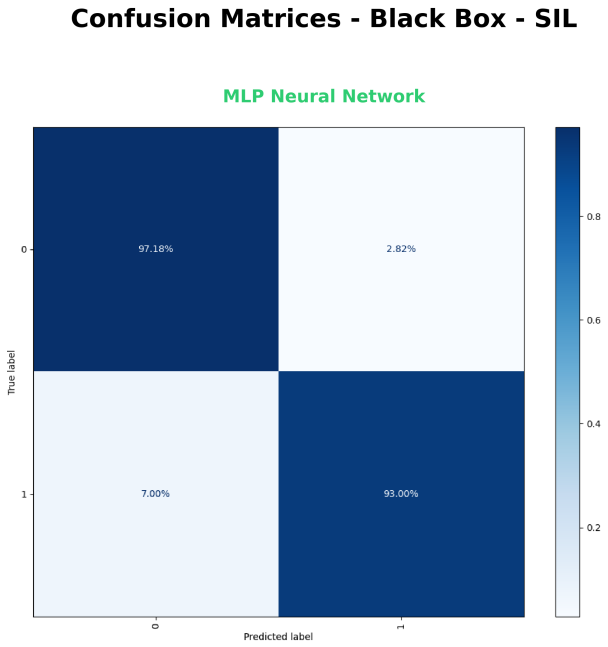


Figure 11: Confusion matrices for the models.

In terms of computational efficiency, the domain-knowledge-based approach (13 variables) offers faster inference times compared to the full PCA projection pipeline (Figure 12), making it more suitable for real-time deployment on UAV hardware.

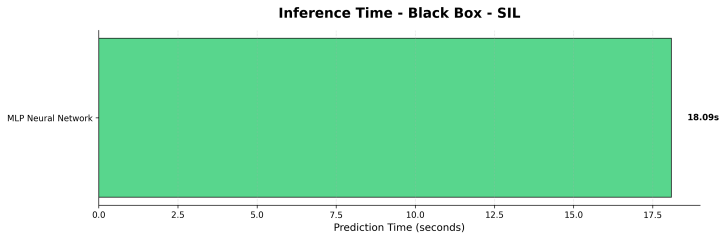


Figure 12: Prediction time comparison across models.

When applying PCA (Figure 13), the F1-Score remains high, confirming that the compressed feature space retains necessary discriminative information.

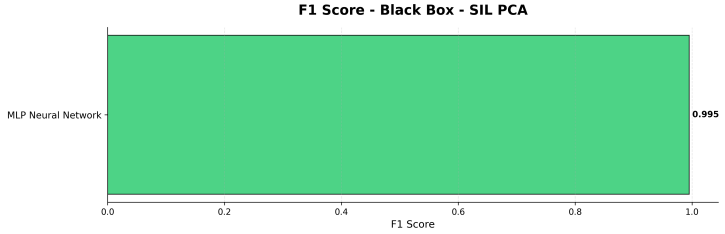


Figure 13: F1 scores using PCA-transformed features.

7 Conclusion

This study presented a complete workflow for binary fault classification in UAVs using the RflyMAD dataset. By transforming sequential logs into static training examples, we enabled the use of standard Black Box classifiers like MLPs. Our results indicate that while Black Box models offer powerful capabilities, careful feature engineering, specifically the removal of experimental artifacts and the use of domain-specific kinematic variables,is crucial for reliable performance. The comparison shows that simpler, domain-aware inputs can achieve results comparable to complex PCA-based pipelines with significantly lower latency.

Furthermore, the successful integration of heterogeneous data sources (SIL, HIL, and Real flights) suggests that the proposed models possess a degree of generalizability essential for practical applications. By training on a diverse set of conditions, the classifiers learn to distinguish genuine fault signatures from environmental noise or simulation artefacts. This robustness is particularly valuable in real-world scenarios where flight conditions are unpredictable and differ from controlled simulation environments.

Ultimately, the validation of the static classification hypothesis opens new avenues for lightweight diagnostics. Since the model does not require the maintenance of temporal windows or complex state history, the computational footprint is minimised. This characteristic is decisive for the deployment of FDI systems on resource-constrained embedded hardware, allowing for real-time monitoring and immediate reaction to critical failures without the latency overhead associated with sequence-based deep learning models.

8 Acknowledgements

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References

Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

Carvalho, T. P. et al. (2019). A systematic literature review of machine learning methods applied to predictive maintenance. *Computers & Industrial Engineering*.

Guyon, I. and Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182.

Hinton, G. E. and Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. *Science*, 313(5786):504–507.

Jolliffe, I. T. (2002). *Principal Component Analysis*. Springer Series in Statistics.

Kotsiantis, S. B., Kanellopoulos, D., and Pintelas, P. E. (2006). Data preprocessing for supervised learning. In *International Journal of Computer Science*.

Lei, Y. et al. (2018). Applications of machine learning to machine fault diagnosis: A review. *Mechanical Systems and Signal Processing*, 113:203–213.

Li, X. et al. (2019). Deep learning-based machinery fault diagnostics using rnns. *Mechanical Systems and Signal Processing*, 106:106–123.

Susto, G. A., Schirru, A., Pampuri, S., McLoone, S., and Beghi, A. (2015). Machine learning for predictive maintenance: A multiple classifier approach. *IEEE Transactions on Industrial Informatics*, 11(3):812–820.

Widodo, A. and Yang, B.-S. (2007). Support vector machine in machine condition monitoring and fault diagnosis. *Mechanical Systems and Signal Processing*, 21(6):2560–2574.

Zhang, Y., Li, Y., Cui, P., and Xu, J. (2019). Deep learning models for industrial predictive maintenance: A comparative study. *Journal of Manufacturing Systems*, 53:20–30.

Zhao, D. et al. (2019). Deep convolutional neural network based planet bearing fault classification. *Computers in Industry*.